

A Project Report on

Machine-Learning Based Image Processing System for Monitoring Road Construction

Submitted in partial fulfillment of the requirements for the award
of the degree of

Bachelor of Engineering

in

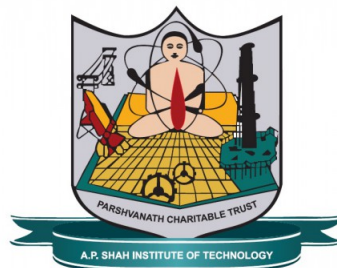
Information Technology

by

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Approval Sheet

This Project Report entitled “*Machine-Learning Based Image Processing System for Monitoring Road Construction*” Submitted by “*Sumit Kanse*”(22104052) is approved for the partial fulfillment of the requirement for the award of the degree of *Bachelor of Engineering* in *Information Technology* from *University of Mumbai*.

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Declaration

I declare that this written submission represents my ideas in my own words and where other's ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

In the ever-expanding field of infrastructure development, efficient monitoring of road construction projects is crucial for ensuring timely completion, cost-effectiveness and quality compliance. Traditional monitoring methods depend on manual inspections by the engineers and supervisors, which are growing number of projects across cities, relying solely on field visits results in delays, inconsistent reporting and limited transparency, ultimately hindering effective project management. To overcome these challenges, we introduces a machine learning based monitoring system designed to automate the analysis of road construction progress using site images. The system validates uploaded images, detect errors and classifies the stage of construction through trained models. It further compares the new submission with an previously stored records to check completed and pending work, providing stakeholders with reliable progress insights. By automatically generating structured reports and visual summaries, the platform ensures accurate, transparent and real-time updates without requiring frequent on-site visit. This project not only reduces manual efforts but also enhances accountability and decision making for urban local bodies, contractors and central agencies. In the future, solution can be expanded to include other infrastructures works such as bridges, flyovers and the utilities, while integrating with IoT and GIS for the real time monitoring. By automating the progress tracking and improving transparency, the system represents a step toward smarter, more efficient and sustainable infrastructure management.

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List of Abbreviations

ML:	Machine Learning
CRUD:	Create, Read, Update, Delete
UAT:	User Acceptance Testing
UI:	User Interface
SQL:	Structured Query Language
VS Code:	Visual Studio Code
API:	Application Programming Interface
UAV:	Unmanned Aerial Vehicle
CNN:	Convolutional Neural Network
IDE:	Integrated Development Environment
RAM:	Random Access Memory
JPG/PNG:	Image file formats
XAMPP:	Cross-Platform Apache, MariaDB, PHP, and Perl
URI:	Uniform Resource Identifier

Chapter 1

Introduction

Road construction is the one of the most critical aspects of the infrastructure development, as it directly impacts connectivity, trade, economic growth, and the quality of life of citizens. The successful execution of the such projects requires close supervision and continuous monitoring to ensure that timelines, budgets, and the quality standards are maintained. Monitoring also helps the identify delays, resource mismanagement, or deviations from the planned process, thereby enabling the corrective actions at the right time. Traditionally, the monitoring of physical progress in the road construction has relied heavily on field inspections by technical experts. Engineers and supervisors visit the project sites periodically, observe the ongoing work, and the records the status in reports. While this provides ground-level visibility, it comes with major drawbacks. The large number of projects spread across cities makes the frequent inspections impractical. Field visits are time-consuming, costly and prone to human error. Moreover, data collected manually often lacks uniformity and transparency, leading to inefficiencies and the possible disputes between contractors, agencies and authorities. The shortcomings of this traditional approach extend beyond more inefficiency, they create a fundamental breakdown in trust and communication. Contractors may face payment delays due to subjective progress validation, while government agencies lack and reliable, objectives data required for effective auditing and future planning. This absence of a single, verifiable 'source of truth' makes accountability difficult to enforce and hinders the ability of urban local bodies (ULBs) to provide citizen with accurate project timelines. As Indian cities continue to expand, the number of road projects being executed simultaneously is increasing rapidly. In such a scenario, depending solely on manual monitoring is not sustainable. There is a pressing need for a scalable, automated solutions that can track construction progress reliably and in near real time. Recent advancements in machine learning and computer vision provides a strong foundation for such a solution. By analyzing image captured from construction sites, machine learning models can automatically determine the type of activity being carried out, identify the stage of progress, and generate systematic reports. This approach eliminated the dependency on frequent site visits, ensures more accurate assessments, and allows agencies to monitor multiple projects simultaneously from a central platform. A machine learning-based monitoring solution would not only improve efficiency and the transparency but also enable proactive decision making. It could flag inconsistencies in uploaded images, validates whether the data corresponds to the correct project site, and compare progress over time. This would help government agencies, contractors and the urban local bodies (ULBs) achieve greater accountability, improved resource utilization and timely project delivery.

1.1 Motivation

The traditional method of monitoring road construction through manual site visits is inefficient, consuming the significant time and the public funds while being prone to human error and delays. As urban areas expand, this reliance on subjective, infrequent inspections becomes unsustainable, leading to inconsistent quality control and a lack of the reliable, real-time data needed for effective project management.

This project is therefore motivated by the urgent need to shift from this reactive model to a proactive, data-driven one. By leveraging machine learning, we can create a smarter framework to automatically analyze site images. The core goal is to provide an accurate and transparent view of construction progress, enabling faster decision making, enhancing accountability and supporting the timely delivery of critical public infrastructure.

1.2 Problem Statement

Monitoring road construction projects in the large cities is a difficult task when done manually. Engineers and supervisors need to visit multiple sites, which takes a lot of time, efforts and cost. Due to limited manpower, inspections are not done regularly and the different people may judge the same work differently, leading to confusion and inconsistent quality. This manual process also reduces transparency, as authorities and citizens cannot easily track the real-time progress of projects. Delays in identifying the mistakes result in rework, increased costs and the wastage of resource. It also creates the disputes between contractors and the government agencies due to a lack of clear proof of progress. To solve these issues, there is a need for an automated and intelligent monitoring system. The proposed solution uses machine learning and image processing to analyze construction site images and identify the current stage of work. Site managers can upload images, and the system generates the accurate progress reports. Additionally, citizen can view project updates, improving transparency and the accountability. This system reduces manual efforts, ensures consistent monitoring and helps in completing the road construction projects efficiently and on time.

1.3 Objectives

- To develop a machine learning system for automated monitoring of road construction progress using uploaded site images.
- To identify the stages of construction through the image analysis, ensuring accurate recognition of different roadwork activities.
- To generate automated progress reports and visual summaries that provide stakeholders with clear, timely and reliable updates.
- To implement a secure user management system where Site Managers/Contractors must be verified and approved by the Admin before accessing system features.
- To enable Site Managers/Contractors to create, update and manage multiple road construction projects with proper details and image uploads.
- To provide Citizens access to view the projects progress, to check the road status and uploads images for monitoring or comparing to check the progress.

1.4 Scope

- Can be extended to other infrastructure projects (buildings, flyovers, waterworks).
- Can be scaled for use in other civil infrastructure projects beyond roads, such as bridges or highways.
- Can be integrated with the IoT and sensors for real time monitoring.
- Can be enhanced with the predictive analytics for accurate completion timelines.
- Can be accessed by project managers, engineers and contractors for transparent status review and communication.
- Can be store long term project data for future maintenance and road quality analysis.

1.5 SDG Mapping and Contribution

- **Industry, Innovation and Infrastructure (SDG 9):** This project supports innovation and sustainable infrastructure development by using artificial intelligence and machine learning to improve the monitoring of the road construction projects. It brings the digital technology into traditional construction processes, helping engineers and authorities track progress more efficiently. By automating image analysis and the progress reporting, the system reduces the human errors, improves quality control and supports the creation of reliable management through data-driven insights and transparency.
- **Decent Work and Economic Growth (SDG 8):** The project contributes to decent work and economic growth by increasing efficiency and reducing manual supervision time in the road construction projects. Through automation, it helps workers focus on the higher-value tasks and reduces the chance of repeated inspection or delays. It promotes accountability among project teams and contractors, ensuring the timely project completion and better use of resources. This leads to cost savings, improved productivity and overall economic growth in the construction sector.
- **Sustainable Cities and Communities (SDG 11):** By improving the accuracy and transparency of road construction monitoring, the project supports the development of the well-planned and sustainable urban infrastructure. It ensures that roads are built on time and maintained properly, enhancing connectivity and safety for communities. The use of ML-based monitoring also minimizes unnecessary site visits and paperwork, promoting eco-friendly and efficient project management. This ultimately helps create smarter, safer and more sustainable cities for future generations.

Chapter 2

Literature Review

A. Omanovi's et al.[1] presented the purpose of the "Progress Monitoring of Road Construction" work, the present work proposes a 'UAV'. A novel method for automatic monitoring of road construction using ortho images and a Convolutional Siamese Network. Instead of checking the videos, the system compares the image taken from the UAV at the different times and detects the changes like new asphalt and changes in the slope, but filters out the changes like shadows, vehicles and vegetation. This work has been implemented on a real construction site in South Korea. In the present work, the full eBEE Plus Drones were used and data processing was done using Pix4D Mapper and modeling using PyT. For GPU Servers, the method has been implemented and has proved to be more reliable compared the pixel method, eliminating false positives. This work aims at the making the road complete, intelligent and the efficient structural monitoring through the use of the drones.

H.-W. Choi et al.[2] focused on a work in the Construction Industry by H.-W. Choi to present the work proposed a discussion on the transformation of the industry across the design, the construction and maintenance phases. It while presenting the definition of the drones and their application, also discusses the importance of the drones and how they are used in the construction industry. In the design, the drones are used for site selection, land survey, mapping and generating maps and 3D models. In the maintenance phase, it helps with the detection of cracks, structural defects and insulation problems using the drones with high resolution cameras, LiDAR and the thermal sensors. The article shown how drones are capable of creating efficiencies, safety, results and cost-effectiveness and the challenges of short flight times, weather dependence and processing complications. In conclusion, the paper shows that as technology advances, the drones are the major importance in the development of smart and sustainable construction.

D. Han et al.[3] works with the title of "Change Detection in unmanned Aerial Vehicle Image for Progress Monitoring of Road Construction" with unmanned Aerial system, an application study from Switzerland, presents a case study on how to apply Unmanned Aerial System (UAS) for monitoring the progress and quality of a commercial building currently under construction in the city of Zurich. It has three phases that automated data collection with the help of drones, processing and generating an ortho image and 3D point cloud and finally analyzing the data where comparisons of the data with Building Information Modeling (BIM) and schedules are needed. Using a DJI Inspire drone and software like Agisoft Metashape, Autodesk Recap 3D and detecting construction deviation. In addition, it also

shows a cost-benefit analysis of the three scenarios that using own drones, the outsourcing the service and the renting a third party service, compared with the traditional manual monitoring. The study found that although in house use of UAS had a slightly higher cost of about 8.5 percent, it provided faster insights and additional visual documentation. The study concludes that the drones can be affective for construction monitoring, though further automation and integration of AI are required to reduce time and processing cost.

Reja et al.[4] introduced the analyzing of various vision-based techniques used in the construction monitoring. The research reviewed multiple methods such as data acquisition, information retrieval, progress estimation and visualization to understand how automation can improve construction tracking. The study highlighted that computer vision can significantly reduce the manual work by automatically analyzing site images and generating progress insights. It also identified key challenges such as alignment issues, multi-modal data fusion and the need for fully automated system. The paper concludes that computer vision-based monitoring has strong potential but requires further improvements for real world implementation.

J.Y. Han et al.[5] explored a comparative assessment of their effectiveness through real-time monitoring of a highway construction project in Indonesia. The research methodology involved using data collected from DJI Phantom UAV and a conventional total station surveying instrument on the Cileunyi-Sumedang-Dawuan Highwat project. The results showed that the UAV method was the highly advantages, generating the detailed 3D photo-grammetric models that were easy for both experts and laymen to understand compared to conventional 2D point data. Quantitative analysis confirmed that UAV method was 35.41 percent faster, 78 percent more economical and the achieved productivity gain of the 94.4 percent without compromising accuracy. The study concludes that the UAV technology provides a high resolutions, real time and the cost effective solution for the construction monitoring, through limitation such as battery life and waterproofing remain.

L. Yingdong et al.[6] discusses on the improving accuracy of construction progress tracking using deep learning techniques. The research introduced an advanced algorithm based on the modern object detection models (such as YOLOv8) to automatically identify and classify different stages of construction from site images. The methodology involve training the model on a large dataset of construction images and evaluating its performance using standard metrics like precision, recall and mean average precision(map). The results showed that the proposed model achieved high accuracy (around 95 percent map) and was capable of detecting construction activities efficiently in real-time scenarios. Compared to traditional manual monitoring methods, the system significantly reduced human effort and improved the speed of progress analysis. The study concludes that deep learning based approaches provide a reliable, accurate and the scalable solution for construction monitoring, although challenges like dataset quality and environmental variation may still affect performance.

R. Zhang et al.[7] implemented the Vision based real time progress tracking and productivity management in building construction progress tracking and productivity management in building construction and focused on improving construction monitoring using computer vision techniques. The research introduced a vision based system that uses cameras and deep learning model to track construction progress and measure productivity in real time. The methodology involved capturing continuous site images and applying image processing and

object detection algorithms to identify ongoing construction activities and compare them with planned schedules. The results showed that system could accurately track progress and provide real-time insights into construction performance, helping in better project management. It also improved productivity monitoring by detecting delays and inefficiencies early. Compared to traditional manual methods, the approach reduced human effort and increased accuracy and speed of analysis. This study concludes that vision-based systems are effective for real-time construction monitoring and productivity management, although challenges such as changing site condition and data quality still need to be addressed.

Pinnamaaraju T. S. Priya et al.[8] worked on the identifying road damages using image processing techniques. The research introduced a deep learning-based approach using Convolutional Neural Networks (CNN) to automatically detect potholes from the road images. The methodology involved collecting a dataset of road images with and without potholes, pre-processing them, and training the model to classify and detect damaged areas accurately. The result showed that the system was able to detect potholes with good accuracy and efficiency, reducing the need for manual road inspections. It also helped in faster identification of road defects, which can support timely maintenance and improve road safety. Compared to traditional inspection methods, the proposed system reduces human effort and operational cost. The study concludes that deep learning based potholes detection is an effective and scalable solution, although factors like lighting conditions and image quality may affect performance.

Sr No.	Author	Year	Contribution	Advantages	Disadvantages	Findings
[1]	Omanovi et al.[1]	2021	UAV-based system using orthoimages and Siamese Network for road construction monitoring.	Detects construction changes accurately and removes noise like shadows and vehicles.	Requires UAV setup, GPU processing, and complex implementation.	More reliable than pixel-based methods with reduced false positives.
[2]	H.-W. Choi et al.[2]	2023	Study on drone applications in construction lifecycle (design, construction, maintenance).	Improves safety, efficiency, mapping, and defect detection.	Limited flight time, weather dependency, and data processing complexity.	Drones are essential for smart and sustainable construction.
[3]	D. Han et al.[3]	2021	UAS-based monitoring system integrating BIM and 3D modeling.	Enables delay tracking, visual documentation, and better insights.	Slightly higher cost (8.5%), requires processing time.	Faster insights and effective monitoring compared to traditional methods.
[4]	Reja et al.[4]	2022	Review of vision-based construction monitoring techniques.	Reduces manual work and automates progress tracking.	Challenges in alignment, multimodal data fusion, and automation.	Strong potential but needs improvement for real-world use.
[5]	J. Y. Han et al.[5]	2024	Comparison of UAV and traditional surveying in highway construction.	35.41% faster, 78% cost-effective, high productivity.	Battery limitations and waterproofing issues.	UAVs provide efficient and accurate real-time monitoring.
[6]	L. Yingdong et al.[6]	2025	Deep learning (YOLOv8) based construction progress detection.	High accuracy (95% mAP), real-time detection, reduced human effort.	Depends on dataset quality and environmental conditions.	Reliable and scalable solution for construction monitoring.
[7]	R. Zhang et al.[7]	2025	Vision-based system for progress tracking and productivity management.	Real-time monitoring, detects delays and inefficiencies.	Affected by site conditions and data quality.	Improves project management and productivity tracking.
[8]	Pinnamraju T. S. Priya et al.[8]	2024	CNN-based pot-hole detection using image processing.	Reduces manual inspection, fast and efficient detection.	Sensitive to lighting and image quality.	Effective solution for road damage detection and maintenance.

Table 2.1: Literature Review Summary in Table Format.

Chapter 3

Project Design

3.1 Existing System

The existing system in a major project report refers to the current method or technology that is already being used to solve a particular problem before the proposed system is introduced. It helps in understanding how the present system works and identifies its strengths and weaknesses.

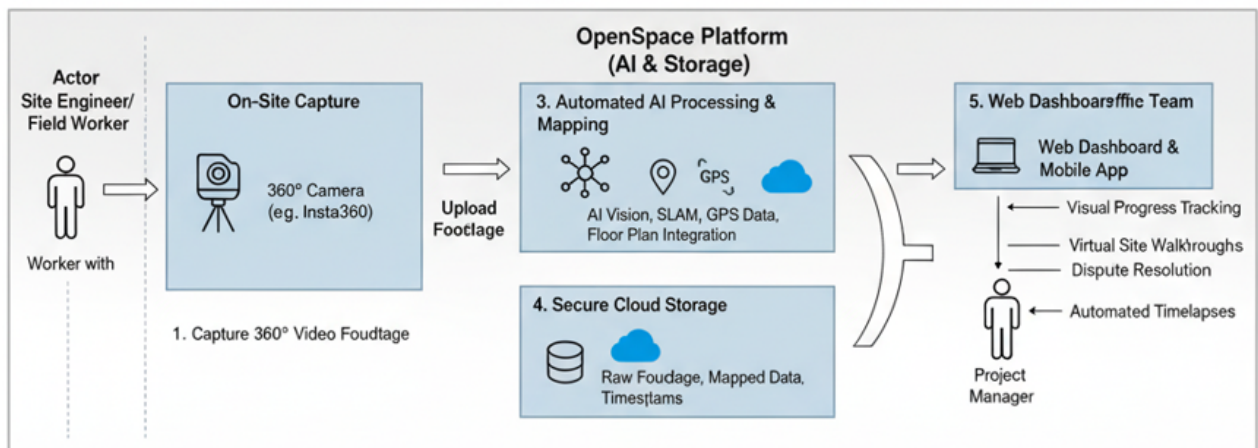


Figure 3.1: Existing System Diagram

In this context, the existing system, OpenSpace, is a leading construction technology platform that specializes in automated 360° photo documentation and progress tracking. Its core concept is to create a comprehensive ‘Google Street View’ for a construction project, allowing teams to virtually monitor progress and inspect site conditions without needing to be physically present. This easy-to-navigate, time-stamped visual record enables remote collaboration, helps verify completed work and provides objective evidence for resolving disputes, ultimately saving time and improving communication among all stakeholders. The working process of OpenSpace is simple and efficient, where a site worker attaches a 360° camera to their helmet and walks around the construction site. As the worker moves, the system captures video footage and its build-in AI automatically maps the data to the project’s

floor plans. This is achieved using advanced computer vision technology and SLAM (Simultaneous Localization and Mapping) algorithms, which analyze video and sensor data to determine the exact position of each image. The processing is performed on a cloud-based backend system, which generates a structured and spatially organized visual database of the construction site.

The system offers several advantages such as reducing the need for frequent site visits, improving the project monitoring, enhancing communication and providing accurate visual documentation of work progress. It also helps in saving time and increasing efficiency in construction management. However, despite these benefits, OpenSpace has certain limitations. It requires specialized equipment like 360° cameras, depends heavily on internet connectivity and cloud processing and may involve high implementation cost. Additionally, it may face environments and limited customization based on specific project needs.

The proposed system is similar to OpenSpace as it also focuses on digital monitoring, efficient data management and improving accessibility to information. Both systems aim to reduce manual work and enhance productivity through technology. However, the proposed system aims to overcome the limitation of OpenSpace by providing a more cost-effective, user-friendly and customized solution with improved accuracy and accessibility.

3.2 Proposed System

The proposed system architecture represents a structured design of the Machine Learning Based Image Processing System for Monitoring Road Construction. It consists of multiple interconnected components that work together to provide efficient monitoring, analysis and reporting of construction progress. In this system, the Core Processing and Machine Learning Layer acts as the critical component, as it performs the main processing and directly influences the system's output and performance. If this layer fails, the overall system will not function effectively.

The architecture is organized into four main layers, starting from the user interface and extending to data storage. The primary users of the system are the Site Manager/Contractors and Admin, who utilize the platform to monitor construction activities and manage project-related data.

The first layer is the Presentation Layer, which provides the user interface for interaction. It includes a web-based dashboard where users can view progress through a tracking interface and generates reports. This layer is designed to be simple, intuitive and user-friendly so that even non-technical users can operate it efficiently.

The second layer is the Application Layer, responsible for handling internal system operations. It consists of a Flask server that processes request received from the frontend. The API Gateway connects various tools and libraries such as PyTorch, OpenCV and Scikit-learn, which are used for ensures smooth communication between the frontend and the core processing components.

The third layer is the Core Processing and Machine Learning Layer, which is the most important part of the system. In this layer, uploaded images are analyzed using machine learning techniques. Initially, an image validation module checks the quality of the input images. Then, a stage classification model identifies the current stage of road construction, such as initial, ongoing or completed. After that, a progress tracking and comparison engine compares current images with previous data to measure development over time. Additionally, an optional GIS component can be integrated to proved location-based analysis for enhanced monitoring.

The final layer is the Data Layer, which manages the data storage. It stores user information, project details, uploaded images and generated reports. A relational database such as MySQL or PostgreSQL is used for structured data, while object storage is used for storing large image files. This ensures secure storage and easy retrieval of the information for future analysis and reporting.

Overall, the proposed system follows a step-by-step workflow where users upload construction images, the backend processes the data, machine learning model analyze the images and the results are stored and displayed to the users. This system reduces manual effort, improves accuracy and provides an efficient and organized approach to monitoring road construction progress.

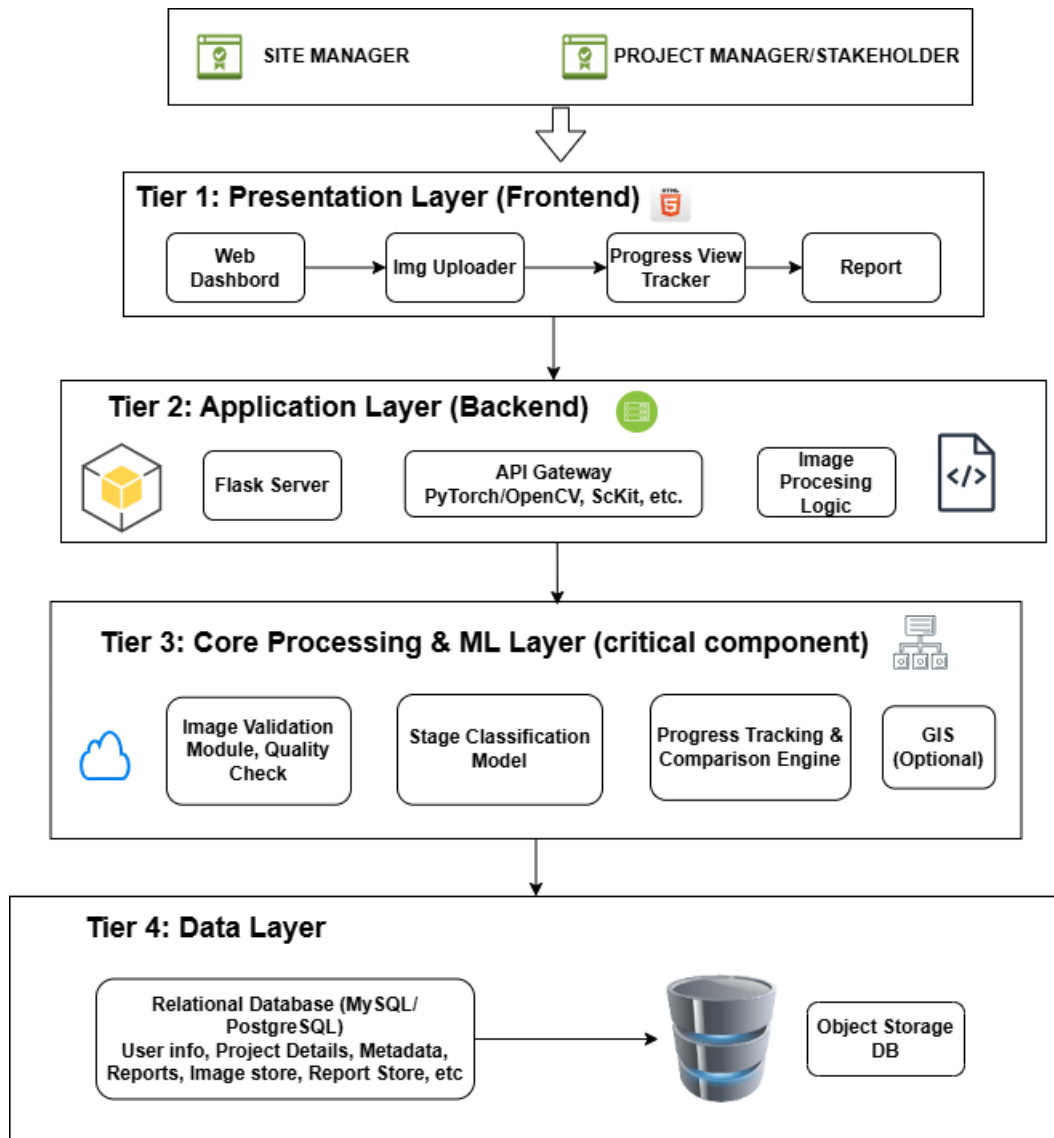


Figure 3.2: Proposed System Architecture

3.2.1 Critical Components of System Architecture

1. User Access and Role Management:

In this system, users are categorized into three main roles: Admin, Site Manager or Contractor and General User (Citizen). The Admin is the responsible for managing user access and approving registration request, when a site Manager or Contractor registers, their account must be verified and approved by the Admin before gaining access to the system. Once approved, Site Manager or Contractors can log in and update the construction details such as location, date, area and upload images of road construction progress. General users can log in to view the updated progress and monitor development. A role-based access to control mechanism ensures that only authorized users can modify data, while other can only view information, maintaining system security and data integrity.

2. Image Upload and Data Management System:

The system allows authorized users only Site Managers/Contractors to upload images along with important project details such as location, data and construction area. This information is stored in a structured format within the database. The uploaded data is organized and maintained efficiently to enable easy retrieval for future analysis and comparison. This component ensures that all construction related data is properly recorded and accessible for monitoring and reporting purposes.

3. Machine Learning Based Image Analysis:

The Machine Learning module is the core component of the system. It analyzed the uploaded construction images using trained datasets and deep learning models. The system first validates the quality of images and then processes them using a classification model to identify the stage of road construction. Additionally, citizen can upload new images to compare with previously stored images. The system evaluates the level of work completed by the analyzing differences between images, providing an estimation of construction progress. This automated analysis reduces manual effort and improves accuracy in monitoring.

4. Progress Tracking and Visualization:

This component provides a clear view of construction progress over time. Based on the analysis results, the system generates progress updates that can be viewed by users through the dashboard. Users can track how much work has been completed and monitor changes in the construction stages. This feature helps stakeholders make informed decision and ensures transparency in project execution.

5. Database and Storage Management:

The system uses a database to store user information, project details and metadata, while image files are stored in a separate storage system. This ensures efficient handling of large data volumes. All data is securely stored and can be accessed when needed for reporting and comparisons. Proper data management enhances system reliability and supports long-term monitoring of construction projects.

3.3 System Diagrams

System diagrams are visual representation of system workflows and user interactions within a system. They provide a clear and structured way to represent the flow of information, the processes that transform data, and how data is stored. System diagrams are important for the system analysis as they help stakeholders understand the functional requirements of the system. This section introduces various diagrams, including UML diagrams such as the Use-Case Diagram, Activity Diagram and Sequence Diagram. These diagram helps in understanding how different components of the system interact with each other. They also illustrate that how data flows through the system and highlight interactions among key users such as Site Manager and Stakeholder, ensuring clarity in system functionality and user roles.

3.3.1 UML Diagram

UML (Unified Modeling Language) Diagrams helps in the visualization and analysis of system behavior, interactions and workflows. This section focuses on the three important diagrams i.e. Activity Diagram, Use-Case Diagram and Sequence Diagram. The Activity Diagram represents the workflow of the system by showing step-by-step processes, including both sequential and parallel activities. The Use-Case Diagram illustrates how different users interact with the system and defines the various functionalities and user roles. The Sequence Diagram shows the flow of messages between system components over time, representing dynamic interactions within the system. Together, these UML Diagrams provides an structured approach to understanding system operations and help ensure efficient system design and implementation.

3.3.2 Activity Diagram

The Activity Diagram represents the workflow of the proposed system, showing the sequence of activities performed from user interaction to final output generation. It illustrates how different component of the system interact to process road construction data efficiently. The process begins when a user accesses the system. If the user is a new Site Manager or Contractor, they must first register by providing required details. The registration request is then sent to the Admin for approval. The Admin reviews the request and either approves or rejects it. If approved, the Site Manager or Contractor gains access to the system, otherwise, access is denied.

Once logged in, the authorized Site Manager or Contractor uploads construction details such location, date, area and images of the construction site. The system then performs image validation to ensure the uploaded data meets the required quality standards.

After validation, the image are processed by the Machine Learning module, which analyze the construction stage using trained models. The system may also compare the uploaded images with previously stored images to determine the level of progress.

Based on the analysis, the system generates the progress updates and reports. These results are stored in database and displayed on the dashboard. Citizens can login to view the projects and reports and also compares the progress with uploading the image.

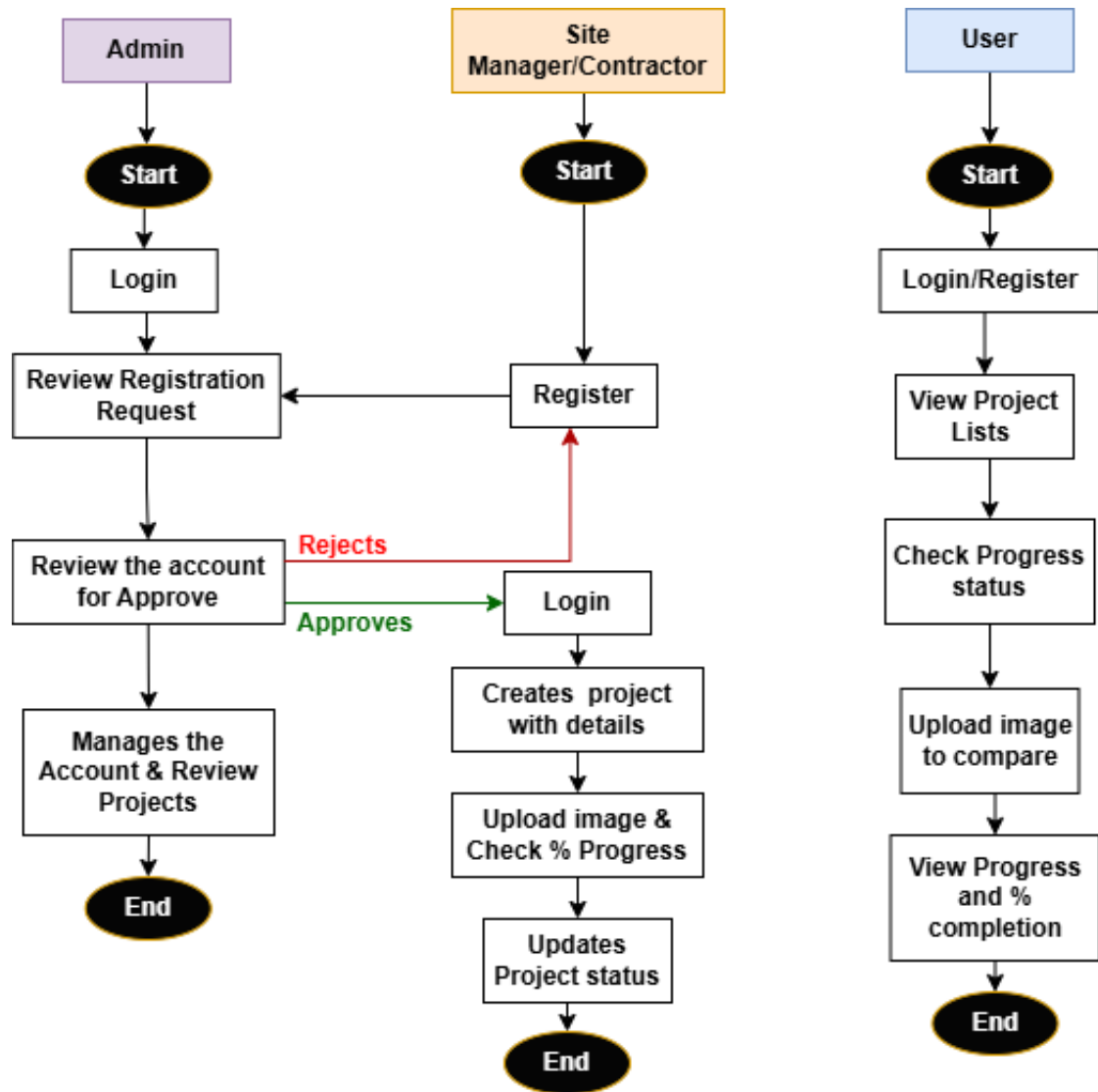


Figure 3.3: Activity Diagram

3.3.3 Use Case Diagram

A Use-Case Diagram is a type of UML Diagram that shows the different user (actors) interact with an system. It helps in the understanding system's functionality by representing the actions that users can perform and how they are connected to the system. It provides a simple overview of user roles and system operations.

In this project, the Use Case Diagram represents the interaction between three main actors: Admin, Site Manager or Contractor and Citizen in the Machine Learning Based Image Processing System for Monitoring Road Construction.

The Admin plays a major role in managing the system. The admin can login, approve or reject registration requests of site managers or contractors and manage user accounts. The admin is also responsible for managing projects and ensuring that only verified Site Manager or Contractor can access and use the system properly.

The Site Manager or Contractor is responsible for handling construction related activities. They can register and login into the system after approval from the admin. Once approved, they can create new road construction projects by entering details such as location, area and timelines. They can upload images of the construction site, update the current status, compare images and check the progress of the projects over time. They can also manage multiple projects.

The Citizen interacts with the system mainly to view and monitor project progress. They can register and login into the system, view available projects, upload images to check the current status of road construction, compare images and check the percentage of the progress.

Overall, the Use Case Diagram shows that how different users interact with the system and perform various actions. It clearly defines user roles and system functionalities, making it easier to understand that how the system works and how responsibilities are divided among users.

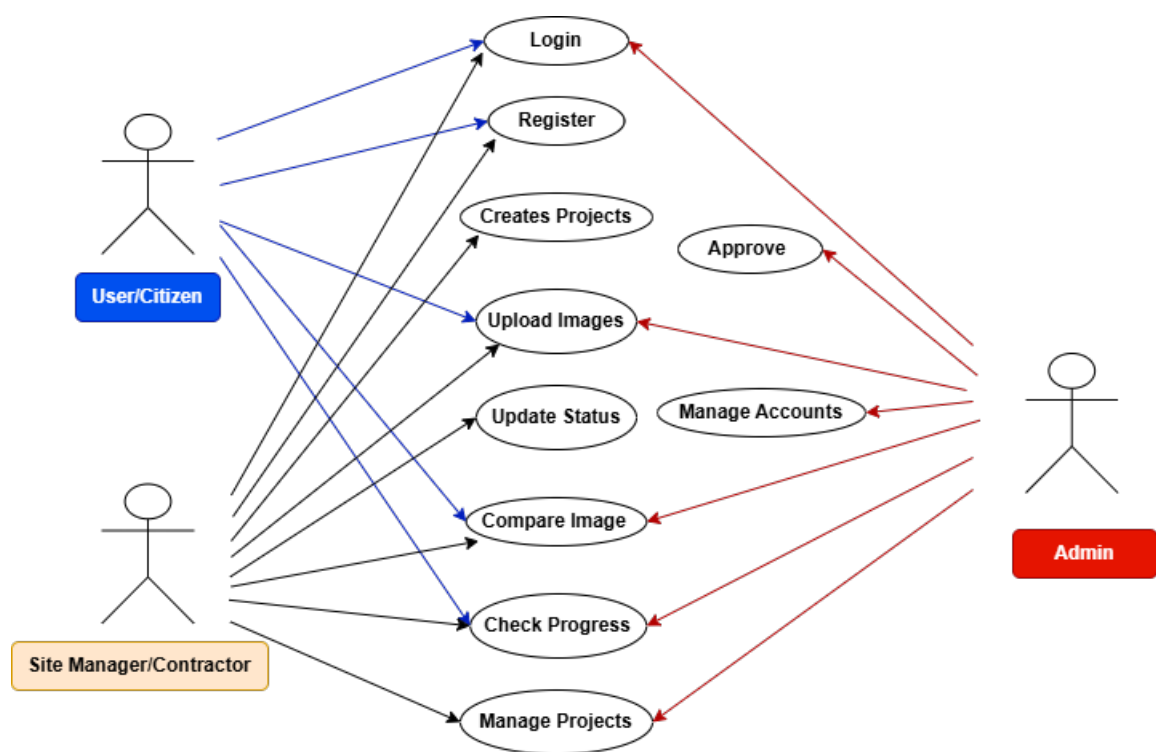


Figure 3.4: Use Case Diagram

3.3.4 Sequence Diagram

A Sequence Diagram shows that how different parts of the system interact with each other step-by-step over time. It represents the flow of messages between users and system components in a proper sequence.

In this project, the sequence diagram includes six main components: Admin, Site Manager, Citizen, System (Backend), Machine Learning Model and Database.

The process starts when the Site Manager or User logs into the system. The login request is sent to the backend system, which verifies the user by checking the database. Once authentication is successful, the system confirms the login.

After logging in, the Site Manager uploads projects images along with the project details. The backend system first validates the input and then stores the project data in the database. Once the data is stored, the system sends the uploaded images to the Machine Learning Model for analysis.

The Machine Learning Model processes the image and detects the construction stage (such as land clearing, earthwork, paving, finishing). The result is sent back to the backend system, which updates the project progress in the database. After successful processing, the system notifies that the upload is completed.

The Citizen can also view the projects status by sending a request to the system. The backend retrieves project information such as progress, timeline and contractor details from the database and displays it to the user.

Additionally, the citizen can upload monitoring images to check the current road condition. These images are stored in the database and the system confirms the upload.

Overall, the Sequence Diagram shows that how data flows between users, backend system, machine learning model and the database in a step-by-step manner. It clearly explains that how the system processes input, performs analysis, stores data and provides results to the users.

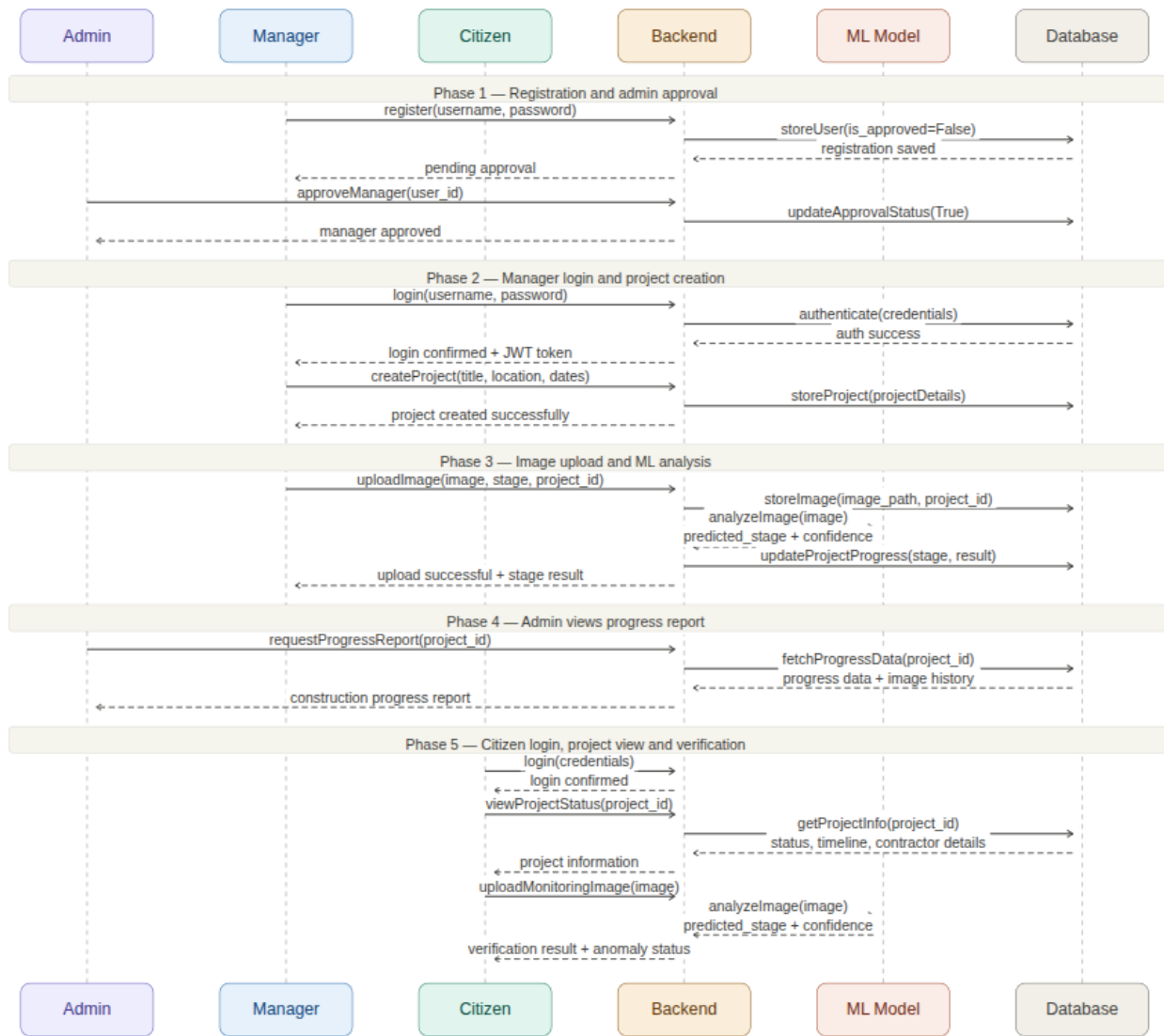


Figure 3.5: Sequence Diagram

Chapter 4

Project Implementation

4.1 Technology Stack

The following technologies were used to develop the Machine Learning Based Image Processing System for Monitoring Road Construction (RCMS).

4.1.1 Frontend

- **React:** React is a JavaScript library used to build all the pages of the RCMS platform such as Login, Dashboard, Project Explorer, and Admin Panel. It updates the page content automatically without full page refresh, making the system fast and smooth.
- **JavaScript:** JavaScript is the programming language used to handle all user actions like button clicks, form submissions, and image uploads. It sends requests to the backend and displays the results on the screen.
- **Vite:** Vite is a build tool used to run and develop the React frontend quickly. It made the development process faster by instantly reflecting code changes without restarting the server.
- **Tailwind CSS:** Tailwind CSS is a styling framework used to design all pages of the system. It helped in creating a clean, modern, and responsive design for buttons, forms, cards, and dashboards without writing separate CSS files.

4.1.2 Backend

- **FastAPI:** FastAPI is a Python framework used to build all the backend APIs of the system. It handles user registration, login, project creation, image upload, and ML model prediction efficiently.
- **SQLite:** SQLite is a lightweight database used to store all system data including user information, project details, image paths, and construction progress records. It is easy to set up and does not require a separate database server.

- **Pandas:** Pandas is a Python library used to organize and manage project data and image records in a structured format. It helped in reading and processing data before storing or using it for reports.
- **Scikit-learn:** Scikit-learn is a machine learning library used for data preprocessing and model evaluation. It helped in preparing image data before passing it to the MobileNetV2 model and measuring prediction accuracy.
- **Pillow:** Pillow is a Python image processing library used to open, resize, and validate construction site images uploaded by users before sending them to the ML model for analysis.

4.1.3 API and Communication

- **REST APIs:** REST APIs are used for communication between the React frontend and FastAPI backend. Every user action like login, project creation, image upload, and progress check is handled through REST API calls.

4.1.4 Server

- **Uvicorn:** Uvicorn is a fast web server used to run the FastAPI backend. It handles multiple user requests at the same time efficiently, making the system quick and responsive.

4.1.5 Data Validation

- **Pydantic:** Pydantic is used to validate all data sent to the backend APIs. It ensures that required fields like username, password, project title, and location are correctly filled before processing any request.

4.1.6 ML Model

- **MobileNetV2 (TensorFlow/Keras):** MobileNetV2 is a deep learning model used to analyze construction site images and identify the current stage of road construction such as Land Clearing, Earthwork, Paving, or Finishing. It was trained using TensorFlow and Keras and gives a predicted stage along with confidence scores for each uploaded image.

4.2 Code Snippets

1. Admin Backend code: This code is used for handling admin operations in the backend system. It retrieves all unapproved site managers from the database. Admin can approve a manager by updating their approval status and saving it. It also includes an API to monitor whether the AI model is trained and ready.

```
from sqlalchemy.orm import Session
from typing import List
import subprocess
import sys
from .. import models, schemas, database, auth
from ..ml_service import get_training_status, reload_model

router = APIRouter(
    prefix="/admin",
    tags=["admin"],
)

@router.get("/pending-managers", response_model=List[schemas.User])
def get_pending_managers(
    current_user: models.User = Depends(auth.get_admin_user),
    db: Session = Depends(database.get_db)
):
    return db.query(models.User).filter(models.User.role == "manager", models.User.is_approved == False)

@router.put("/approve-manager/{user_id}", response_model=schemas.User)
def approve_manager(
    user_id: int,
    current_user: models.User = Depends(auth.get_admin_user),
    db: Session = Depends(database.get_db)
):
    user = db.query(models.User).filter(models.User.id == user_id).first()
    if not user:
        raise HTTPException(status_code=404, detail="User not found")
    user.is_approved = True
    db.commit()
    db.refresh(user)
    return user

@router.get("/training-status")
def check_training_status(
    current_user: models.User = Depends(auth.get_admin_user),
):
    """Returns image counts per stage, whether minimum is met, and if model is trained."""
    return get_training_status()
```

Figure 4.1: Admin Backend code

2. FastAPI Backend Setup Code: This code initializes the FastAPI application for the project. It creates database table and connects all the route modules (auth, admin, project, citizen). CORS middleware is added to allow frontend-backend communication. It also sets up an folder and route for serving uploaded images as static files.

```
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from .database import engine, Base
from .routers import auth, projects, citizen, admin

from fastapi.staticfiles import StaticFiles
import os

Base.metadata.create_all(bind=engine)

app = FastAPI(title="Road Construction Monitoring System")

# Create static directory if not exists
os.makedirs("backend/static/uploads", exist_ok=True)

# Mount static files
app.mount("/static", StaticFiles(directory="backend/static"), name="static")

# CORS middleware to allow frontend communication
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"], # In prod, specify frontend URL
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

app.include_router(auth.router)
app.include_router(admin.router)
app.include_router(projects.router)
app.include_router(citizen.router)

@app.get("/")
def read_root():
    return {"message": "Welcome to RCMS API"}
```

Figure 4.2: FastAPI Backend Setup Code

3. React Frontend Code: This code manages the frontend routing using React. It controls access based on the user roles (Admin, Site manager, Citizen). Private routes protects the pages from unauthorized users. It connects the different pages like login, dashboard, etc.

```
import React, { useContext } from 'react';
import { Routes, Route, Navigate } from 'react-router-dom';
import Navbar from './components/Navbar';
import { AuthContext } from './context/AuthContext';

// Pages
import Home from './pages/Home';
import Login from './pages/Login';
import Register from './pages/Register';
import AdminDashboard from './pages/AdminDashboard';
import ManagerDashboard from './pages/ManagerDashboard';
import CitizenDashboard from './pages/CitizenDashboard';

const PrivateRoute = ({ children, role }) => {
  const { user, loading } = useContext(AuthContext);

  if (loading) return (
    <div style={{ display: 'flex', height: '80vh', alignItems: 'center', justifyContent: 'center' }}>
      <div className="spinner"></div>
    </div>
  );

  if (!user) {
    return <Navigate to="/login" />;
  }

  if (role && user.role !== role) {
    return <Navigate to="/" />;
  }

  return children;
};

const App = () => {
  return (
    <div className="app">
      <Navbar />
      <main>
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/login" element={<Login />} />
          <Route path="/register" element={<Register />} />

          <Route path="/admin" element={
            <PrivateRoute role="admin">
              <AdminDashboard />
            </PrivateRoute>
          } />

          <Route path="/manager" element={
            <PrivateRoute role="manager">
              <ManagerDashboard />
            </PrivateRoute>
          } />

          <Route path="/citizen" element={
            <PrivateRoute role="citizen">
              <CitizenDashboard />
            </PrivateRoute>
          } />
        </Routes>
      </main>
    </div>
  );
};
```

Figure 4.3: React Frontend Code

4. Model Training - MobileNetV2 Code: This code builds a deep learning model using MobileNetV2 for image classification. The pre-trained base model is frozen to reuse learned features. Additional layers like pooling, dense and dropout are added for better learning. The model is compiled with Adam optimizer, loss function and accuracy metric for training.

```
# — Training —  
  
def build_model(num_classes):  
    """Build MobileNetV2 transfer learning model."""  
    base_model = MobileNetV2(  
        weights='imagenet',  
        include_top=False,  
        input_shape=(IMG_SIZE[0], IMG_SIZE[1], 3)  
    )  
    # Freeze base model layers  
    base_model.trainable = False  
  
    x = base_model.output  
    x = GlobalAveragePooling2D()(x)  
    x = Dense(128, activation='relu')(x)  
    x = Dropout(0.3)(x)  
    predictions = Dense(num_classes, activation='softmax')(x)  
  
    model = Model(inputs=base_model.input, outputs=predictions)  
    model.compile(  
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),  
        loss='sparse_categorical_crossentropy',  
        metrics=['accuracy']  
    )  
    return model
```

Figure 4.4: Model Training - MobileNetV2 Code

5. Prediction and Comparison Code: This code is used to predict the construction stage from an image input. It first tries to use the trained AI model for accurate prediction. If the model is unavailable, it switches to a histogram based comparisons method. It returns the predicted stage, confidence percentages and the method used for prediction.

```
def predict_construction_stage(image_path: str) -> dict:
    """
    Analyzes the image and returns the predicted construction stage.
    Uses trained model if available, otherwise falls back to histogram comparis

    Returns dict with:
        - predicted_stage: str
        - confidence_scores: dict[stage -> percentage]
        - method: "trained_model" or "histogram_fallback"
    """
    # Try trained model first
    stage, scores = _predict_with_model(image_path)
    if stage is not None:
        return {
            "predicted_stage": stage,
            "confidence_scores": scores,
            "method": "trained_model"
        }

    # Fallback to histogram
    stage, scores = _predict_with_histogram(image_path)
    return {
        "predicted_stage": stage,
        "confidence_scores": scores,
        "method": "histogram_fallback"
    }

def compare_images(image_path_1: str, image_path_2: str) -> dict:
```

Figure 4.5: Prediction and Comparison Code

4.3 Steps to Access the System

To ensure smooth functionality and easy usage, different users (Admin, Site Manager/Contractor and Citizen) follows specific steps to access and use the Road Construction Monitoring System (RCMS).

4.3.1 Admin Access:

1. Login to the System:

- Open the RCMS portal.
- Enter the Admin credentials.
- Click on the Login button.
- If credentials are correct, the Admin dashboard is displayed.

2. Manage Site Manager Requests:

- Navigate to pending manager requests.
- Approve or reject Site Manager registration.

3. Manage Projects:

- View all ongoing and completed projects.
- Monitor project progress and activities.

4. Monitor System and Model:

- Check training status of the Machine Learning Model.
- Ensure the system is functioning properly.

4.3.2 Site Manager / Contractor Access:

1. Register and Login:

- Register into the system with required details.
- Wait for Admin approval.
- Login after approval using credentials.

2. Create New Project:

- Navigate to the Management Console.
- Enter the project details (title, description, location, dates, etc.).
- Click on the Deploy Project.

3. Manage Projects:

- View created projects.
- Update project stages.

- Uploads site images as proof.

4. **Track Progress:**

- Compare images over time.
- Check progress percentage generated by system.

4.3.3 **Citizen Access:**

1. **Register and Login:**

- Register with basic details.
- Login using credentials.

2. **View Projects:**

- Browse available road construction projects created by Site Managers.
- View project details and updates.

3. **Upload Images:**

- Upload images of construction site for compare.
- Help in monitoring real time progress.

4. **Check Progress:**

- Compare images updated image with the current uploading images.
- View system generated construction stage and progress percentage.

4.4 Timeline SEM VIII

This is a Gantt Chart for a week project to create a “Machines Learning Based Image Processing System for Monitoring Road Construction”. It visually lays out the project’s timeline for SEM VII and SEM VIII. The chart lists specific tasks like ‘Literature Review’ and ‘Project Design’, showing their start and end dates.

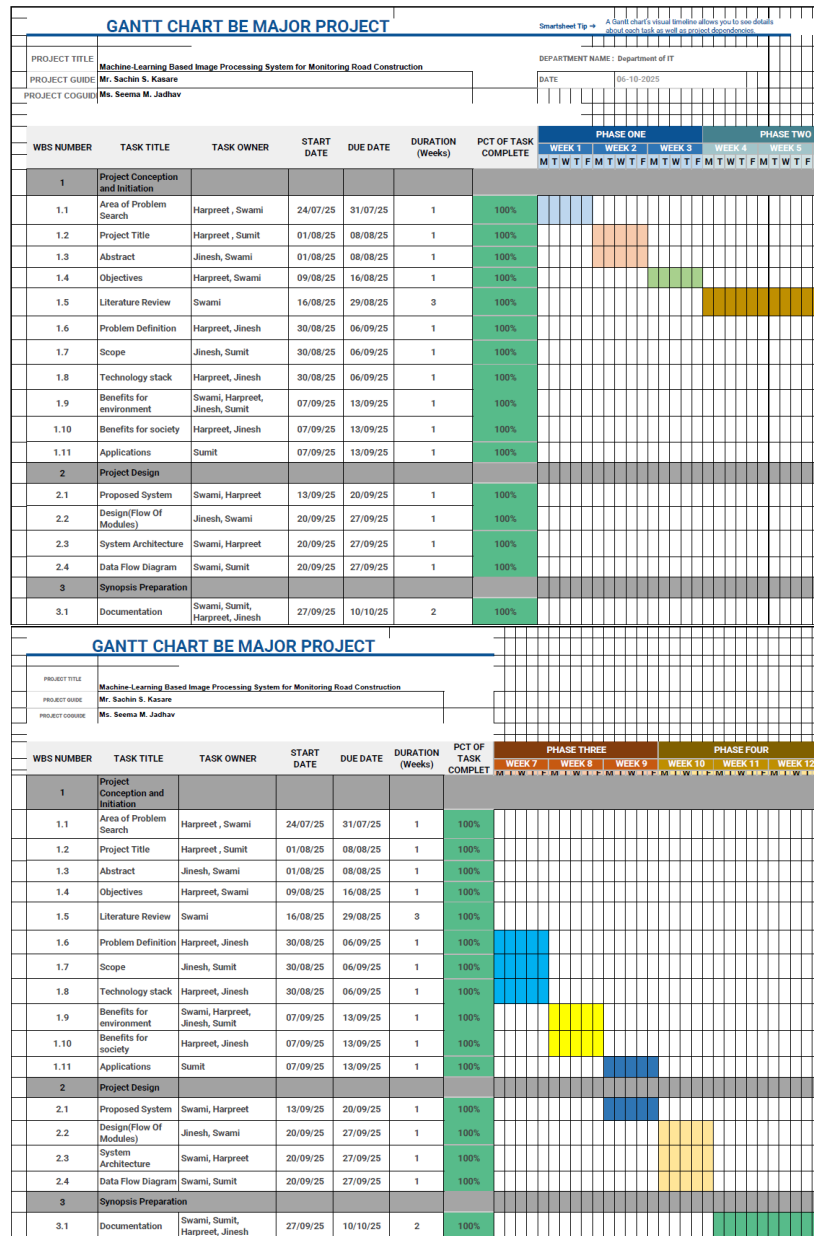


Figure 4.6: Timeline SEM VII

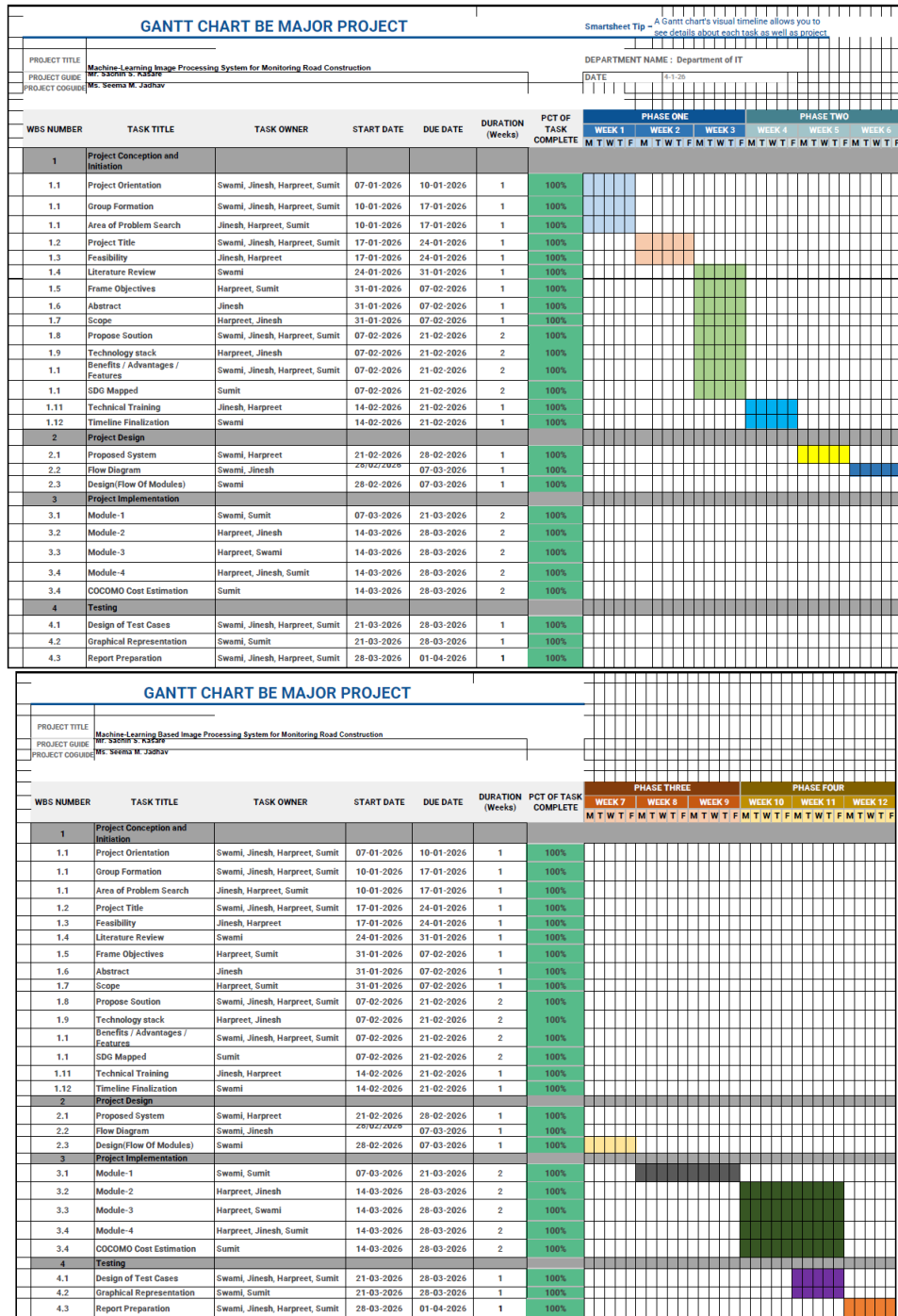


Figure 4.7: Timeline SEM VIII

4.5 Cost Estimation using COCOMO-II Model

Step 1: Identify the Objects

- **Screens (UI Forms):**
 - Project Dashboard
 - Image Upload Page
 - Upload History Page
 - Report Page
 - Report Store Page
- **Reports:**
 - Progress Timeline Report
 - ML analysis Report
- **3GL Component:**
 - ML stage prediction module
 - Database CRUD Handler
 - File storage engine
 - User authentication handler

Step 2: Classify Complexity and Assign Weights

1. Screen (5): Simple – 2 (Weight 1); Medium – 2 (Weight 2); Difficult – 1 (Weight 3)
2. Reports (2): Simple – 1 (Weight 2); Medium – 1 (Weight 5); Difficult – 0
3. 3GL Components (4): Simple – 1 (Weight 10); Medium – 2 (Weight 15); Difficult – 1 (Weight 20)

Screen Points: $(2 \times 1) + (2 \times 2) + (1 \times 3) = 2 + 4 + 3 = 9$

Report Points: $(1 \times 2) + (1 \times 5) = 2 + 5 = 7$

Components Points: $(1 \times 10) + (2 \times 15) + (1 \times 20) = 10 + 30 + 20 = 60$

Step 3: Calculate UOP (Unadjusted Object Points)

$$\text{UOP} = 9 + 7 + 60 = 76$$

Step 4: Adjust for Reuse (%RE = 15%)

$$\text{NOP (New Object Points)} = 76 \times (1 - 0.15) = 64.6 \approx 65$$

Step 5: Effort Estimation

Productivity = 13 OP/person-month

Total Effort = $65/13 = 5$ person-months

Time (with 4 developers) $5/4 = 1.25$ months

Step 6: Cost Estimation

Developer Cost = $1.25 \text{ months} \times 30,000 \text{ Rs./month} \times 4 = 1,50,000 \text{ Rs.}$

System & Platform Cost = 30,000 Rs.

Miscellaneous Costs = 10,000 Rs.

Final Cost Table

Paid Software/API Cost	–	0
Developer Cost	$1.25 \times 30,000 \times 4$	1,50,000
System & Platform Cost	–	30,000
Additional H/W	–	0
Miscellaneous Costs	–	10,000
Total Estimated Cost		1,90,000

Table 4.1: COCOMO-II Cost Estimation for ML-Based Image Processing System for Monitoring Road Construction

Chapter 5

Testing

5.1 Software Testing

- **Unit Testing:** Individual functions/methods (such as the ML stage prediction and database CRUD operations) were manually tested for image checking and resizing validated using sample images.
- **Integration Testing:** The system was tested end-to-end by uploading images through the frontend, ensuring the workflow from upload seamlessly. Image deletion, project report generation and timeline tracking validated together.
- **System Testing:** The full application was deployed in a test environment (localhost with MySQL), testing real user scenarios: project creation, bulk uploads, full project lifecycle and edge cases (invalid image, empty uploads, rapid navigation).
- **User Acceptance Testing (UAT):** Prototype shown to the target users (Site Managers), who validated the usability of dashboard, upload, history and reporting features.
- **Error Testing:** Upload corrupt/unsupported image format, system displays proper error. Attempted action on already deleted item i.e. user is safely redirected.

5.2 Functional Testing

Functional testing checks if all parts of the software work as expected. It tests features like creating projects, uploading images and generating reports to make sure they do what they are supposed to. This ensures the application runs smoothly and meets the user's needs.

Test Case	Description	Expected Result	Status
TC-01	Site Manager/Contractor can register with verification request	Registration request sent to Admin for approval	Pass
TC-02	Admin can approve or reject Site Manager/Contractor registration	Approved Contractor can login, rejected Contractor must re-register	Pass
TC-03	Citizen/Admin/Site Manager can login to the system	Successful login redirects to dashboard	Pass
TC-04	Site Manager/Contractor can create new road construction project	Project details saved and displayed in project list	Pass
TC-05	Site Manager/Contractor can upload road images	Image stored successfully and linked to project	Pass
TC-06	Uploaded image is validated (quality check)	Low quality images are rejected with message	Pass
TC-07	ML model analyzes image and detects construction stage	Correct stage and progress percentage displayed	Pass
TC-08	System tracks and updates project progress over time	Progress percentage updates correctly	Pass
TC-09	User can upload image to check road condition	System analyzes and shows current stage	Pass
TC-10	User can upload image to check road condition	System analyzes and shows current stage	Pass
TC-11	Admin can manage projects and user accounts	Changes reflected correctly in system	Pass
TC-12	User can generate and downfall project report	Report generated with correct details	Pass

Table 5.1: Functional Testing Results for Road Construction Monitoring System

Chapter 6

Result and Discussions

The Machine Learning Based Monitoring System significantly improved the oversight of the road construction projects. By automatically analyzing site images to determine the construction stages and track progress, the system provided accurate and timely updates. This increased operational efficiency, enhanced transparency for all stakeholders, and contributed to the timely completion of projects.

1. Home Page: This homepage presents an ML Based Image Processing Monitoring Road Construction. It helps track real time progress using images and machine learning. Users can explore projects or access a manager dashboard for detailed insights. The goal is to improve transparency, accuracy and efficiency in project monitoring.

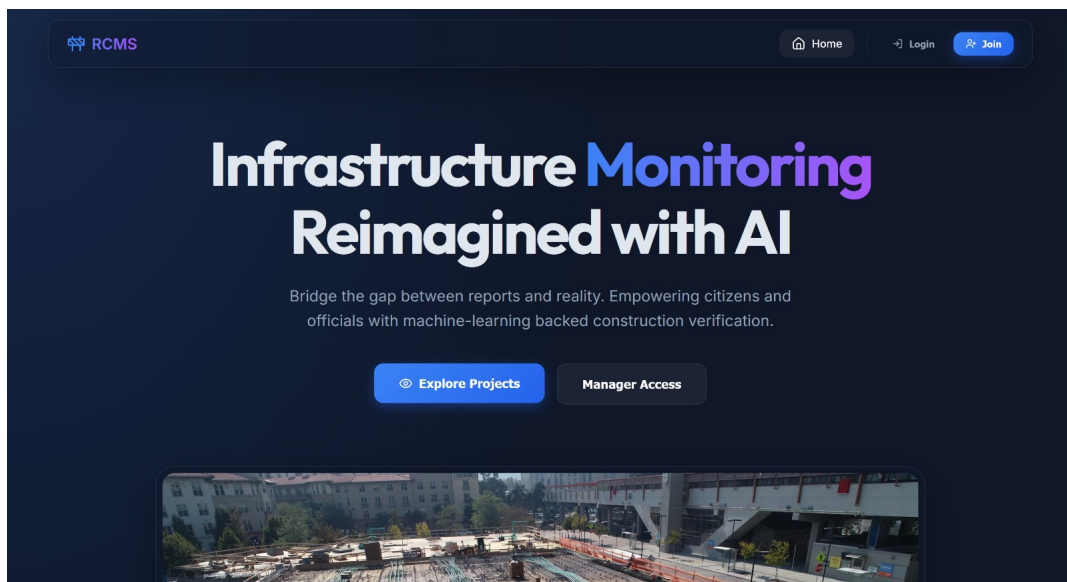


Figure 6.1: Home Page

2. Login Page: This page allows users to securely login to the system using their username and password. It provides access to the dashboard where users can monitor road construction progress. The system ensures safe and authorized entry for Site Managers. It supports efficient access to real time project updates and reports.

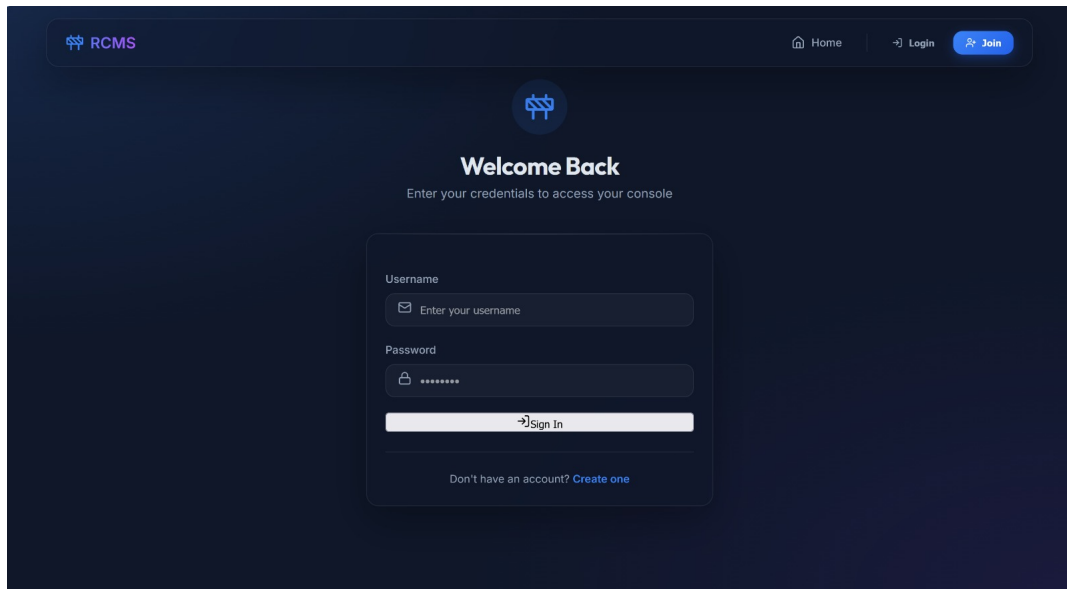


Figure 6.2: Login Page

3. Signup Page: This page allows new users to create an account in the system. Users can register as a Citizen or Site Manager to access different features. It collects basic details and enables users to start the monitoring road construction system. The system promotes transparency by allowing more users to view and track the project progress.

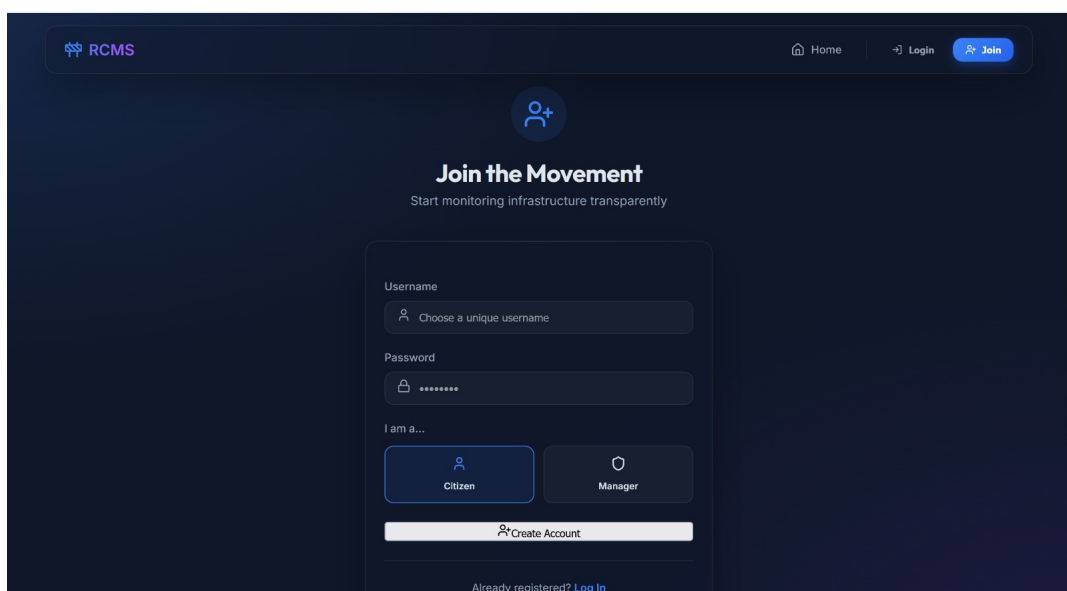


Figure 6.3: Signup Page

4. Admin Page: This page allows the Admin to monitor and control the entire system. It shows system status, pending approvals and manages user access. The Admin can train the machine learning model using construction stage images. It helps ensure accurate monitoring and smooth management of road construction projects.

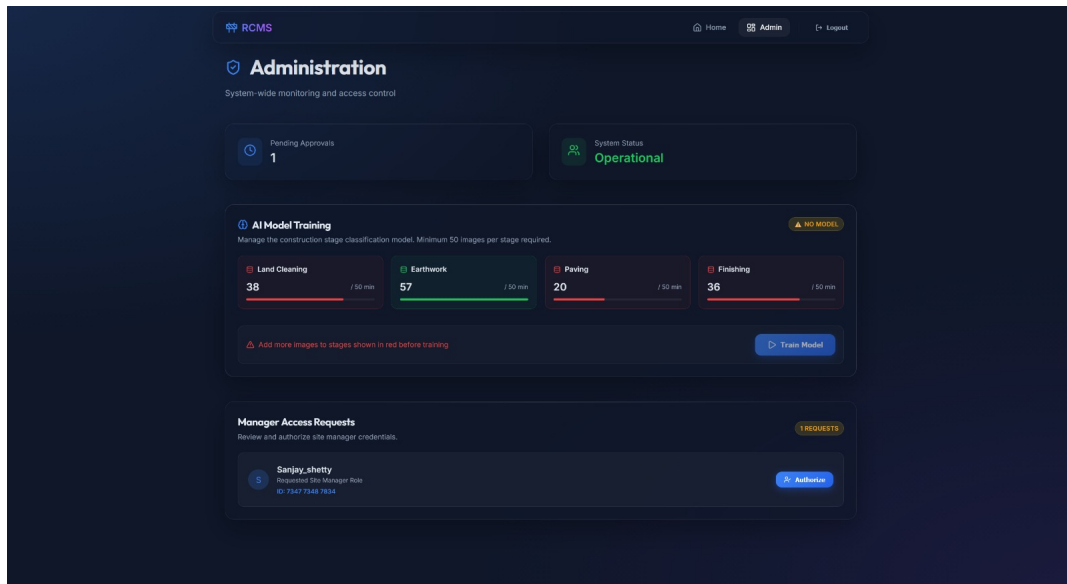


Figure 6.4: Admin Page

5. Admin Dashboard: This page is an Admin dashboard for managing the Site Manager requests. It shows a section for pending Site Manager approvals where Admin can review request. There is an approve button to quickly accept the request.

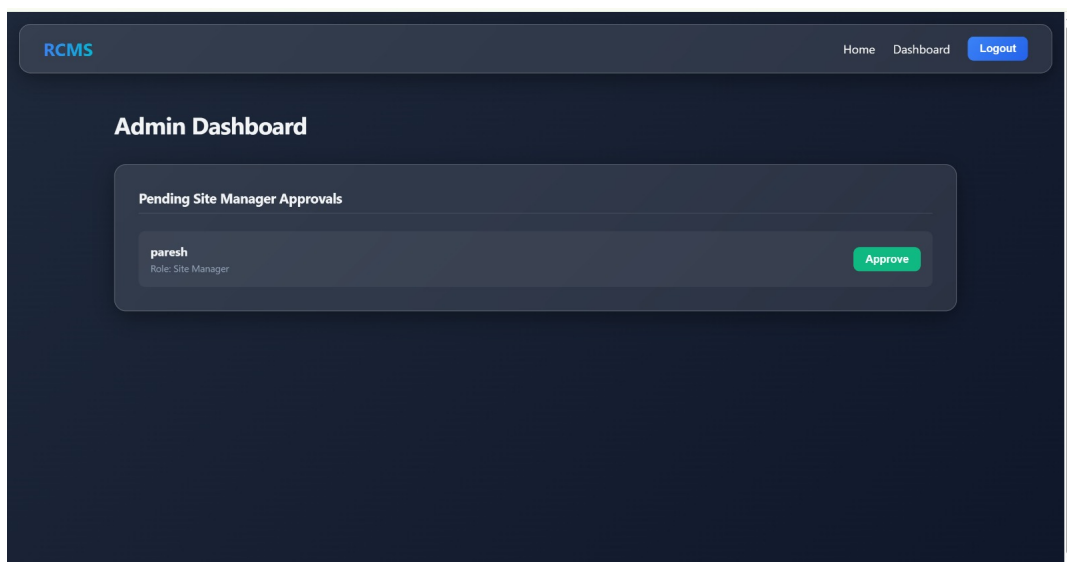


Figure 6.5: Admin Dashboard Page

6. Project Explorer Page for Citizen: This page is the Project Explorer where Citizen can see all available projects for Citizens. Each project card shows an image, name and short description which is created by Site Manager or Contractor. Citizen can search project by name, location. There is an button to view full project details. This helps Citizen to easily track and monitor road construction progress.

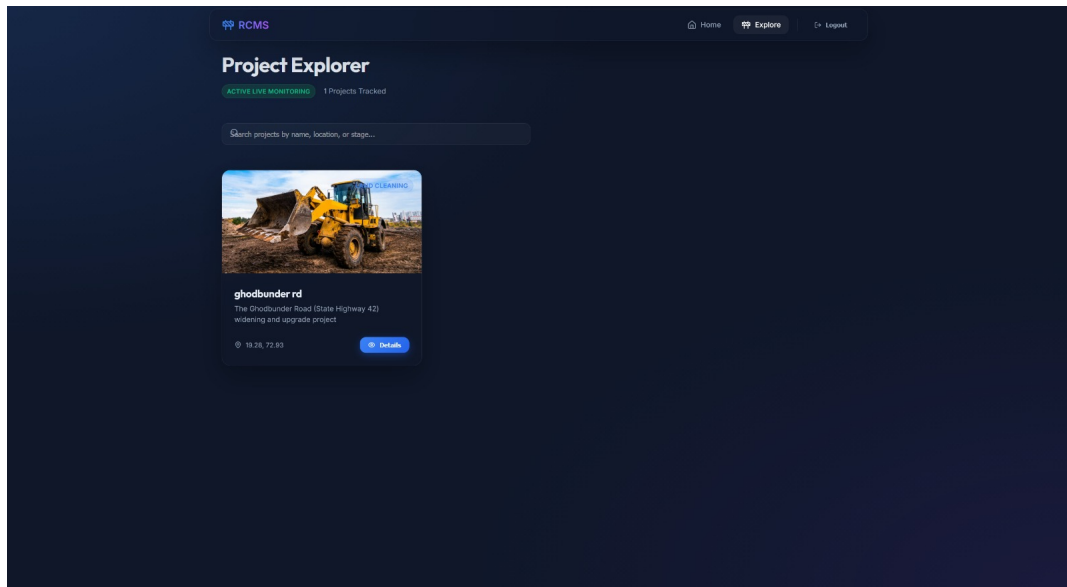


Figure 6.6: Project Explorer Page

7. Project Details Page: This page shows detailed information about a selected road construction project. It includes the project name, description, contractor details and timeline. There is a section for an official updates showing the latest work progress. Citizen can upload a real time site image to verify the actual work progress.

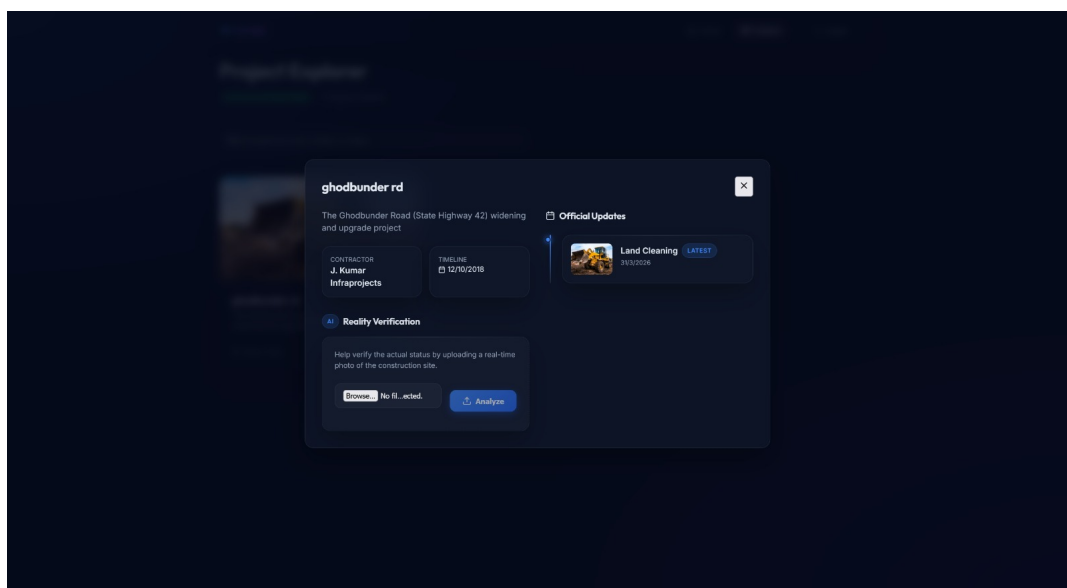


Figure 6.7: Project Details Page

8. Result Analysis Page: This screen shows verification result of the uploaded construction image. It compares the official update with the work stage. A progress breakdown with the percentage is displayed for different activities. If there is any mismatch, the system highlights it as an anomaly for transparency.

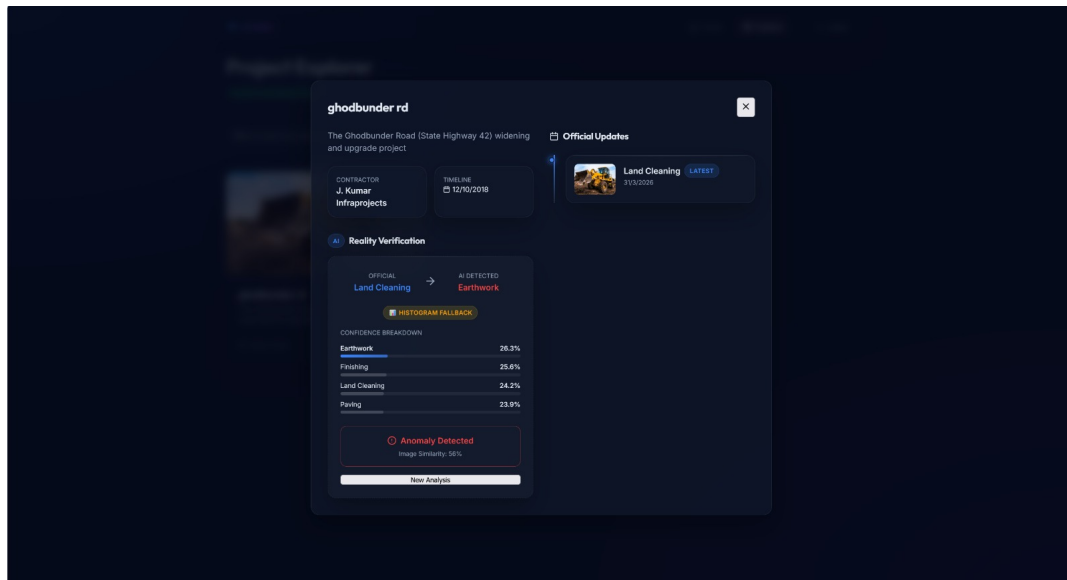


Figure 6.8: Result Analysis Page

9. Management Console Page for Site Manager/Contractors: This page is the Management Console dashboard for only Site Manager or Contractor. It shows a list of projects they have created and managing. There is an Post Status Update button to upload new project updates for the projects. Only Site Manager/Contractors has access to create a new project using the New Project button.

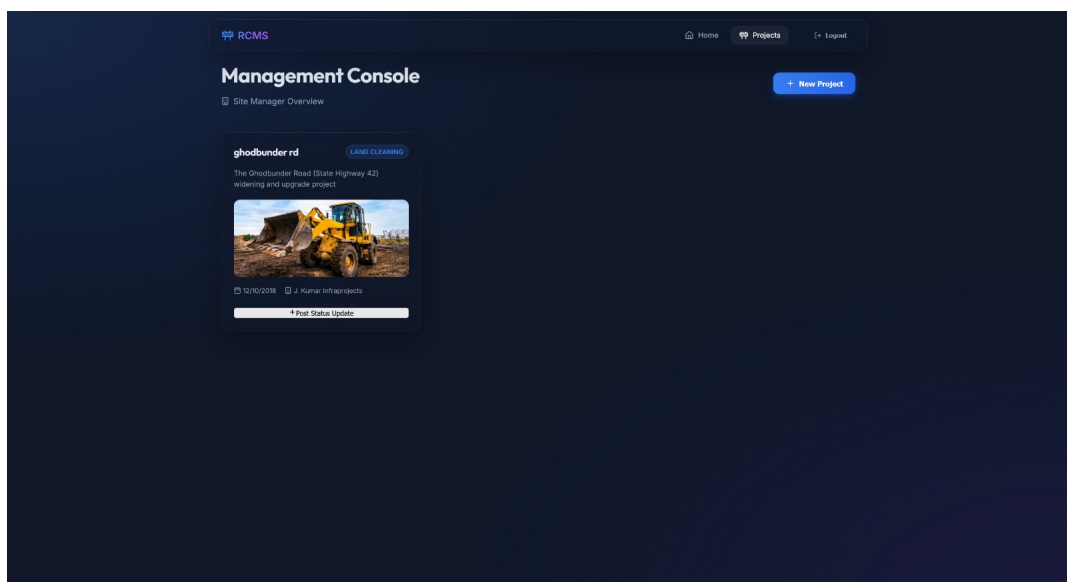


Figure 6.9: Management Console Page

10. Project Creation Page: This page is the Management Console used by the Site Manager or Contractor. It allows them to create a new road construction project. They enter details like project title, description, location (latitude and longitude) and Contractor name. They can also set the start date and planned end date of the project.

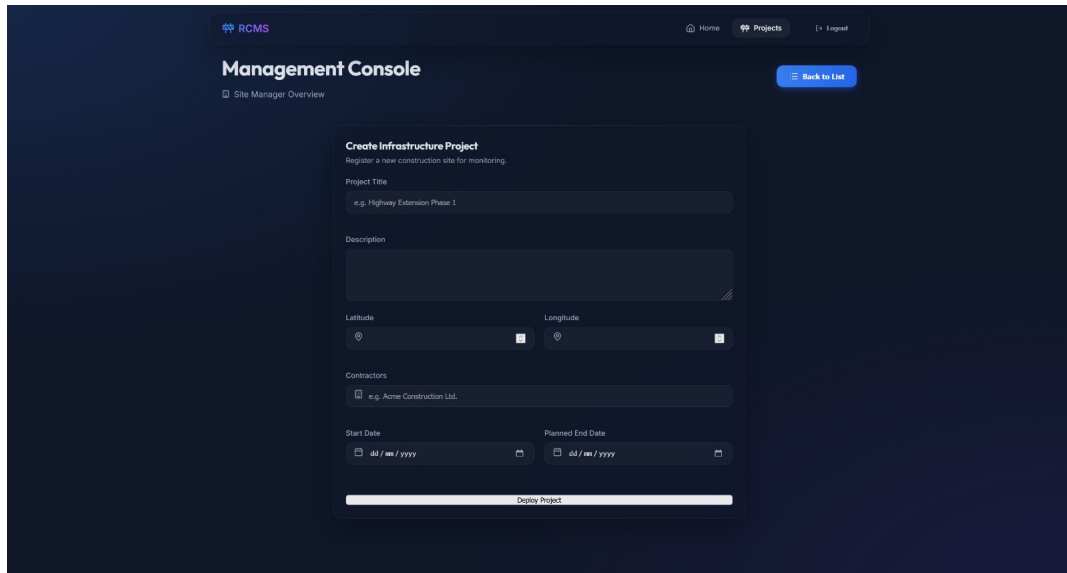
The screenshot shows a web application interface for creating a new infrastructure project. At the top, there's a navigation bar with 'RCMS' on the left and 'Home', 'Projects', and 'Logout' on the right. Below the navigation bar, the main heading is 'Management Console' with a 'Site Manager Overview' link and a 'Back to List' button. The central form is titled 'Create Infrastructure Project' with a sub-header 'Register a new construction site for monitoring.' The form fields include: 'Project Title' (text input with placeholder 'e.g. Highway Extension Phase 1'), 'Description' (text area), 'Latitude' and 'Longitude' (each with a location pin icon and a square selection icon), 'Contractors' (text input with placeholder 'e.g. Acme Construction Ltd.'), 'Start Date' (calendar icon and placeholder 'dd / mm / yyyy'), and 'Planned End Date' (calendar icon and placeholder 'dd / mm / yyyy'). At the bottom of the form is a 'Deploy Project' button.

Figure 6.10: Project Creation Page

11. Project Updates: This page shows the Post Update form for a project by Site Manager or Contractor. It only allow the Site Manager or Contractor to update the current construction stage. They can select stafe (like earthwork, finishing, etc) from an drop-down. They also need to upload an image as proof of work done. After filling the details, they can click publish update to submit the progress.

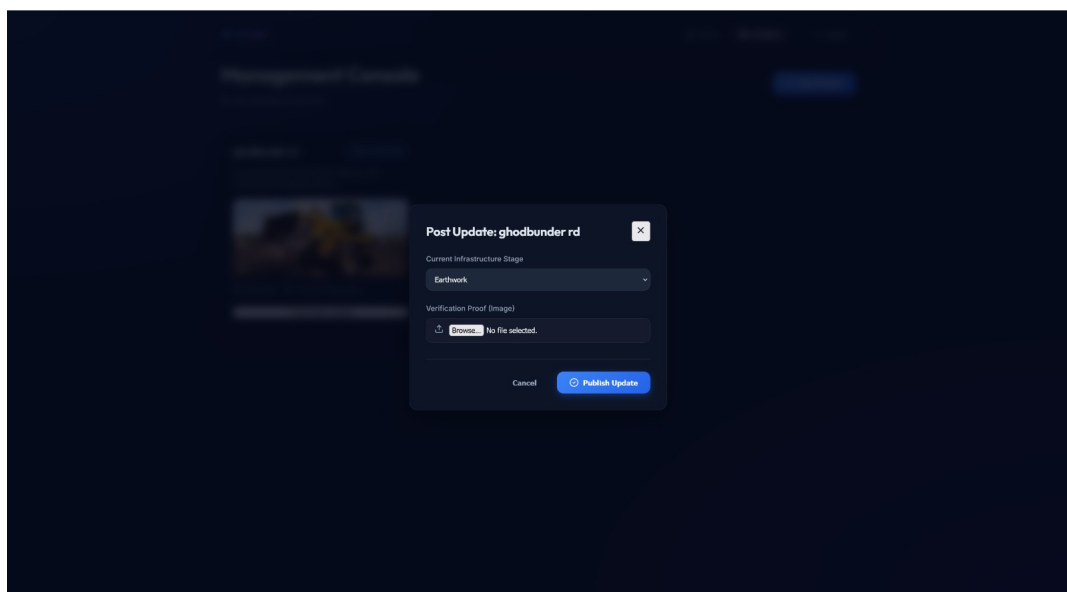
The screenshot shows a modal form titled 'Post Update: ghodbunder rd' with a close button. The form contains: 'Current Infrastructure Stage' (a dropdown menu with 'Earthwork' selected), 'Verification Proof (Image)' (a file upload section with a 'Browse' button and the text 'No file selected.'), and two buttons at the bottom: 'Cancel' and 'Publish Update'.

Figure 6.11: Project Update Status Page

Chapter 7

Conclusion

This project titled Machine Learning Based Image Processing System for Monitoring Road Construction was developed to improve the way road construction projects were monitored and managed. The system provided a digital platform where Admin, Site Manager and Citizen were able to interact and manage project activities efficiently. It allowed Site Managers to create projects, upload images and update the progress of the road construction in a structured way. The use of Machine Learning helped in analyzing the uploaded images and identifying the stage of the construction, which made progress tracking more accurate and reliable. The system maintained the time stamped image records, which helped in the clear understanding of the condition of the construction site at different stages. This made monitoring easier and reduced the need for frequent visits. It also helped in resolving disputes by providing clear visual proof of completed work. The platform improved transparency, as all stakeholders were able to view the real time project updates and progress details. The system also helped in identifying issues such as poor construction practices at an early stage, which allowed timely corrections and reduced future repair costs. This contributed to better quality construction and efficient use of resources. The Admin module ensured proper verification of the users and smooth management of projects, making the system more secure and reliable. Overall, the project achieved its goal of providing an efficient, user-friendly and cost-effective solution for monitoring road construction. It improved communication, enhanced accuracy and reduced manual effort in project management. Over time, the system can become more intelligent and accurate as more data is collected, helping in better decision making and planning. This approach supports the development of better infrastructure, ensures proper use of funds and contributes to safer and more reliable roads for the community.

Chapter 8

Future Scope

In the future, the system can be improved by adding more advanced features to make road monitoring more effective and intelligent. One possible improvement is road classification visualization, where different types of the road such as asphalt, concrete and gravel can be automatically identified and displayed with the images. This will help users easily understand the type and condition of the road. The system can also include analytical insights that provide detailed information about the accuracy of predictions. Metric like confidence level can help user understand how reliable the results and improve trust in the system. Another improvement is adding more specification details for each project. Information such as location, contractor name, contractor photo and construction equipments can be stored and displayed along with the images. This will make system more informative and help in better documentation and tracking. The Admin panel can be enhanced to provide better control and management features. Admin can analyze data, compare project performance and generates more detailed reports for decision making and planning. In the future, the system can also support real-time monitoring using live image capture from cameras or drones. Integration with GPS and GIS can provide location based tracking of road construction. The machine learning model can be improved over time by using more data which will increase more accuracy and better performance.

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Appendices

This section details the setup process for developing, running, requirement and maintaining the Road Construction Monitoring System using Python, Flask, React, MySQL, Visual Studio Code (VS Code), etc.

Appendix-A: Setting Up the Road Construction Monitoring Project

1. Prerequisites Ensure the following are installed:

- Python 3.8 or newer
- pip (Python package manager)
- Visual Studio Code (VS Code)
- MySQL Server (optional)
- Git (optional)
- XAMPP (optional)

2. Intall Visual Studio Code

Download from:

<https://code.visualstudio.com/>

3. Recommended Extensions

- Python
- Pylance
- SQL Tools
- Prettier/Black Formatter
- MagicPython

4. Project Setup a. Clone or Download Project

```
git clone <repository-link>
```

b. Create Virtual Environment

```
python -m venv venv  
venv\Scripts\activate
```

c. Install Dependencies

```
pip install -r requirement.txt
```

Libraries Used:

- Flask
- Tensorflow
- OpenCV-Python
- Numpy
- Pandas
- PymySQL
- Pillow

d. Configure Database(if used)

```
CREATE DATABASE road_monitor;
```

5. Running the Application

```
python app.py
```

Open in browser

```
http://127.0.0.1:5000
```

6. Notes

- Uploaded Images → static/uploads
- Processed Image → static/processed
- Model Files → .h5

Appendix-B: Dataset Preparation

- Collect the images of Construction stages:
 - Land Cleaning
 - Earthwork
 - Paving
 - Finishing
- Minimum requirement: 50 images per class for better accuracy.

```
dataset/  
  land_cleaning/  
  earthwork/  
  paving/  
  finishing/
```

Ensure Images are clear and properly labeled.

Appendix-C: Model Training

- Load dataset
 - Preprocess image
- ```
python train_model.py

model.hs
```
- Stage Prediction
  - Confidence Score

## Appendix-D: Image Processing and Analysis

- User uploads image
- System processes image
- Model predicts stage
- Displays:
  - Stage
  - Confidence Percentage
  - Comparison Result
  - If mismatch occurs:
- Shows **Anomaly Detected**

## Appendix-E: System Workflow

1. User logs into the system
2. Admin approves managers
3. Manager creates projects
4. Manager uploads updates
5. User upload verification image
6. System compares and analyzes
7. Displays result with accuracy

## Appendix-F: Project Features

- Project creation and management
- Role based login system
- Image upload and verification
- ML based stage detection
- Confidence score generation
- Anomaly Detection system
- Dashboard for Monitoring

## Appendix-G: File Strucutre

```
project/
|-- static/
| |--uploads/
| |--processed/
|-- templates/
|-- models/
|-- app.py
|-- train_model.py
|-- requirement.txt
```

## **Appendix-H: System Requirements**

### **Hardware**

- Minimum 8 GB RAM
- Intel i5 or higher

### **Software**

- Python 3.x
- VS code
- Required libraries

## **Appendix-I: Tools and Technologies**

- Python
- Machine Learning
- OpenCV
- TensorFlow/Keras
- Web Interface
- Database (optional)

## **Appendix-J: Key Capabilities**

- Real-time based monitoring
- ML based image classification
- Progress tracking
- Transparency in construction
- Automated anomaly detection
- Calculates progress reports